Project Brief Documentation

Nusantara Car Motor Sales Marketing Analysis (January 2021 to September 2026 dataset)

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Project description	
Project	Sales analysis for Nusantara Car Motor, a Data Analyst portfolio project
Business	A multi-brand car sales network in Indonesia: 32 dealers in 6 regions, 13 brands, 62 vehicle variants, three customer segments (Retail, Corporate, Fleet)
Data	Four tables query within BigQuery platform: customers 10,690 rows, vehicles 62, dealers 32, transactions 11,939
Period	1 January 2021 to 27 September 2026
Main output	Cleaned data views, 21 prioritised insights, a dashboard and written recommendations

1. Project Background and Purpose

1.a Background

Nusantara Car Motor sells passenger cars through a network of 32 dealers (Main Dealer, Authorized Dealer and Satellite Showroom) across six regions of Indonesia. Customers categorized into three segments: Retail (individual buyers), and Corporate and Fleet (companies, where a Fleet order averages about nine cars). From January 2021 to September 2026 the company recorded 11,939 transactions.

The sales data shows a business that grew fast and is now slowing. Yearly revenue rose from IDR 454 billion in 2021 to IDR 1,279 billion in 2025, but yearly growth fell from +65% in 2022 to +4.4% in 2025. About 9.1% of the value of booked sales is lost to cancelled or returned orders. The business therefore needs one trusted view of which segments, customers, dealers, regions and vehicles drive revenue, and where revenue is lost.

The raw data cannot be used for this directly. It contains duplicate customer records, many spellings of the same city (149 spellings for 36 cities), different phone formats and missing values, so it must be inspected and cleaned before any figure can be trusted.

1.b Purpose

- Business purpose: turn raw sales records into clear answers on where revenue comes from, where it leaks and where the company can grow.
- Analytical purpose: show a complete and repeatable workflow: inspect, cleanse, analyse, visualise and recommend, with every step documented.

2. Project Background and Purpose

Goal

Shared Nusantara Car Motor a trusted, data-backed picture of its sales performance from 2021 to 2026, with practical recommendations to restart growth, reduce revenue leakage and strengthen the dealer network.

Objectives

#	Objective	Key questions
1	Prepare reliable data	Are dataset for each tables clean, consistent and valid?
2	Measure sales performance	How fast is revenue growing, and in which months do sales peak?
3	Understand customers	Which segments and customers matter most? Does growth come from new or repeat customers? How do customers pay?
4	Evaluate the dealer network	Which dealers and regions perform best, and where is demand without a dealer?
5	Analyse products and profit	Which vehicles and brands earn the most, and do discounts increase sales?
6	Find leakage and channel value	How much revenue is lost to cancellations and returns, and which lead sources matter?
7	Communicate the results	Can management understand the findings quickly and act on them?

3. Scope of Work

In scope

- The dataset consist of four source tables: customers, vehicles, dealers and transactions (BigQuery dataset_extraction).
- The period 1 January 2021 to 27 September 2026. Year-on-year comparisons use 2021 to 2025 because 2026 is incomplete.
- Data inspection, cleansing and validation, with every cleaning rule documented.
- Descriptive and diagnostic analysis of 21 prioritised insights, selected from 40 candidate questions.
- Data visualization and written recommendations.

Out of scope

- External data such as competitors, market share or economic indicators.
- Live data pipelines or automated refresh.

Deliverables

Deliverable	Description
Project Brief Documentation	This document: background, goals, scope, process and tools.
Data Inspection and Exploration document	Findings before cleansing: duplicates, NULL values, non-standard values, outliers and validation checks, with queries.
Data cleansing views and CTE documentation	Views clean_customers, clean_vehicles, clean_dealers and clean_transactions, with a step-by-step explanation of each CTE.
Analysis queries and results	validated BigQuery queries and result insight. Also using PowerBI for data visualization
Final insights and recommendations report	Findings, recommendations and limitations in business language.

4. Data Analysis Process

Step	Activity	Output
1. Understand	Review the dataset, define the business problem and the questions to answer.	Project Brief (this document)
2. Load	Load Dataset on the CSV files into BigQuery with schema auto-detection.	Four raw tables in dataset_extraction
3. Inspect	Check duplicates, NULL values, non-standard values, outliers and cross-field rules (for example delivery date after transaction date).	Data Inspection document and queries
4. Cleanse	Build CTE-based views: standardise city names and phone numbers, handle NULL values, and reconcile row counts.	Four clean_ views and CTE documentation
5. Plan	Write business question then correlate with potential insights and test each on the cleaned data.	Organized for insights
6. Analyse	Write and validate SQL queries for insight,	Queries and result tables
7. Visualise	Build the data visualization and matrix from the validated result tables.	Data Visualization(Images)
8. Recommend	Turn the findings into recommendations and state the limitations.	Final report

5. Tools will be Used

Tool	Purpose in this project
Google BigQuery	Data storage, inspection, cleansing views and analysis queries, using CTEs, joins, window functions (ROW_NUMBER, SUM OVER, QUALIFY) and aggregate functions.
Google Cloud project	Project project-nusantara-car with dataset dataset_extraction, holding the raw tables and clean views.
CSV source files	customers.csv, vehicles.csv, dealers.csv and transactions.csv, the original data.
Power BI (Desktop)	Data visualization(chart, matrix and table)
Microsoft Word and PDF	Project documentation and the guiding-insights report.
Claude (AI assistant)	Reviewing SQL, cross-checking figures and drafting documentation.